

ALLISON DZUBAK

✉ allisondzubak@gmail.com
🌐 allisondzubak.com
🔗 allison-dzubak

in allisondzubak
📞 515-710-8699
📍 Portland, ME

Making order out of chaos

Versatile data scientist | Ph.D. in computational chemistry with a track record of turning complex, unstructured problems into efficient, data-driven solutions. Combines technical expertise in algorithm design and automation with experience leading projects, mentoring teams, and translating insights across technical and non-technical audiences. Driven by curiosity and impact, with a collaborative and adaptable approach well-suited to dynamic, mission-driven work.

SKILLS

➤ **Data Analysis & Coding**
Python, SQL, Command line, Git, API tools & data integration, ETL, Jupyter, Mathematica, Excel

📊 **Data Visualization & Dashboarding**
Tableau, Streamlit, plotly, seaborn, matplotlib

☁ **Cloud Computing**
AWS, Google Colab

⚙ **Machine Learning & AI**
Dimensionality Reduction, Feature Engineering, Clustering, Regression, Ensemble Methods, Deep Transfer Learning

🌀 **Web Design**
HTML, CSS, Bootstrap

💻 **High-Performance Computing**
Large-scale quantum chemical simulation job management

⚙ **Algorithm Development**
Monte Carlo Methods, Coarse Graining, Optimization Techniques

✍ **Technical Writing**
Grant proposals, Software documentation, Research highlight slide decks, Academic journal publications

👥 **Collaboration & Communication**
Microsoft 365, LaTeX, Digital interactive whiteboards, Video creation, Slack, Trello

👤 **Training & Mentoring**
Universal design for learning, scaffolding, mental health first aid

EXPERIENCE

Independent Data Scientist

📅 Mar 2023 - Present

📍 Remote

- Supported data migration project for Maine Needs local nonprofit by cleaning and transforming 5,000+ donor records, reducing manual import errors and saving 40 hours in human time spent on task
- Contracted by Scale AI to craft advanced chemistry, math, and logic questions to test LLM reasoning; selected to mentor peers based on quality contributions
- Designed and built a solar energy dashboard with automated 15-minute API updates, interactive visualizations, and ROI analysis

Assistant Professor of Computational Chemistry

Bowdoin College

📅 July 2019 - Feb 2023

📍 Brunswick, ME

- Balanced competing research, teaching, and service demands resulting in a successful reappointment towards tenure
- Developed quantum chemical models, resulting in 14 peer-reviewed publications to date and significant advancements in computational methodologies
- Led a research lab with a \$40K annual budget, guiding 4 undergraduate students in advanced computational projects
- Communicated technical findings by generating visualizations of data, having presented at 20+ conferences and workshops in 5 countries
- Assessed the feasibility and merit of proposed research, resulting in 1 faculty candidate hire and approximately \$270K/year of faculty funding
- Managed and facilitated inclusive learning environments, teaching approximately 150 students and enhancing classroom experiences through smart technology

Postdoctoral Research Fellow - Materials Science and Technology

Oak Ridge National Lab

📅 Feb 2016 - Feb 2019

📍 Oak Ridge, TN

- Secured 1.7 million core hours on Titan supercomputer and 100K core hours on EOS for proposed materials theory research, driving new project exploration
- Executed quantum Monte Carlo simulations on 10 transition metals, quantified key errors, prioritized critical approximations, published in J. Chem. Phys. and supported grant renewal

Graduate Research Assistant - Department of Chemistry

University of Minnesota

📅 2009 - 2015

📍 Minneapolis, MN

- Presented collaborative research to DOE program managers 5 times over 3 years contributing to the \$30 million renewal of the DOE carbon capture center
- Defended Ph.D. research on metal-organic framework computational modeling, receiving the American Chemical Society Chemical Computing Group research excellence award and the department annual award for doctoral thesis excellence

EDUCATION & CERTIFICATIONS

- Applied Data Science Program: Leveraging AI for Effective Decision-Making [Massachusetts Institute of Technology](#) (2023 - 2024)
- Ph.D. in Computational Chemistry [University of Minnesota](#) (2009 - 2015)
- B.A. in Chemistry [University of Minnesota](#) (2005 - 2009)